ModelIt!

Teacher's Guide

The Mountain That Used to Be an Ocean

How Seashells End Up on Mountaintops Millions of Years Later

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-ESS1-1, 4-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-ESS1-1, 4-ESS2-1

Identify evidence from patterns in rock formations and fossils in rock layers to support an explanation for changes in a landscape over time. Make observations and measurements to identify materials based on their properties.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a five-component model showing how plate movement and sediment deposition interact through compression and uplift to determine fossil elevation over geological time.
Disciplinary Core Idea	**ESS1.C The History of Planet Earth / ESS2.A Earth Materials and Systems** — Local, regional, and global patterns of rock formations reveal changes over time due to Earth forces such as tectonic plate movement. Fossils provide evidence of organisms that lived long ago.
Crosscutting Concept	**Cause and Effect** — Students trace a chain of cause and effect from plate movement to compression to uplift to fossil elevation, demonstrating how slow, persistent forces create dramatic results over millions of years.

CAST Testing Alignment

MULTIPLE CHOICE:

Scientists find fossils of ancient sea creatures in rock layers near the top of the Himalayan mountains. Which best explains how those fossils got there?

CONSTRUCTED RESPONSE:

Using what you know about tectonic plates, compression, and uplift, explain how a fossil of a sea creature could end up on a mountain that is 5 miles above the ocean. Use the words 'plate movement,' 'compression,' and 'uplift' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

One of the most astonishing facts in science is that the tallest mountains on Earth were once at the bottom of the ocean. The evidence is literally written in stone — marine fossils found at elevations of over 25,000 feet. Understanding how this happened requires thinking about Earth processes that operate over millions of years.

Earth's Moving Puzzle

Earth's outer layer (lithosphere) is broken into about 15 major tectonic plates that float on the hot, partially molten rock beneath (asthenosphere). These plates move very slowly — typically 2-10 centimeters per year, about as fast as your fingernails grow. But over millions of years, this slow movement has rearranged entire continents, opened and closed oceans, and built the world's greatest mountain ranges.

Ocean Floors Become Mountains

The Himalayan mountains are the best example of ocean floor becoming mountaintop. About 200 million years ago, the Indian Plate was a separate landmass far south of Asia. Between India and Asia lay the Tethys Sea, a vast ocean. For millions of years, sea creatures lived and died in this ocean, their shells settling to the bottom and becoming part of the sediment layers. Starting about 50 million years ago, the Indian Plate collided with the Asian Plate. The enormous compression force began folding and pushing up the ocean floor.

Compression, Folding, and Uplift

When two continental plates collide, neither can sink beneath the other (unlike ocean plates, which are denser). Instead, the collision crumples and folds the rock layers like pushing two pieces of paper together on a desk — they buckle upward. This folding and uplift is what creates mountains. The Himalayas are STILL growing today — the Indian Plate continues to push into Asia at about 5 cm per year, and Mount Everest grows about 4 mm taller each year.

Fossils as Evidence

The fossils found in Himalayan rock are the definitive proof that this rock was once ocean floor. Scientists have found fossils of ammonites (ancient shelled sea creatures), crinoids (sea lilies), and marine brachiopods at elevations above 25,000 feet on Mount Everest. The limestone summit of Everest is made of compressed ocean sediment. These fossils could not have formed at that elevation — they MUST have formed at sea level and been carried up by tectonic forces.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Plate Movement Speed and Sediment Deposit Rate as external components (geological conditions) and Compression Force, Rock Uplift, and Fossil Elevation as internal components that respond to those conditions over geological time.

E **Establish Relationships**

Students determine that Plate Movement Speed positively affects Compression Force, Compression Force positively affects Rock Uplift, Sediment Deposit Rate positively affects Fossil Elevation, and Rock Uplift positively affects Fossil Elevation — a system with converging inputs.

V **Visualize & Model**

Students build a five-component model showing how plate collision creates compression that drives uplift, while sediment deposits determine how many fossil-bearing layers are available to be lifted.

E **Evaluate Outcomes**

Students run four scenarios (fast/thick, slow/thin, no movement, fast/thin) to observe how both inputs contribute to fossil elevation and what happens when one or both are absent.

R **Revise & Extend**

Students add erosion rate to explore why some mountains with fossils are shorter than expected — erosion removes material from the top even as uplift pushes from below.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"I have a mystery for you. Scientists found SEASHELLS on top of Mount Everest — five miles above the ocean. How did ocean creatures get to the top of the tallest mountain on Earth?"*

Watch For

Show a real photo of marine fossils found on Himalayan peaks. Let students react and discuss initial ideas.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to model something that took FIFTY MILLION YEARS to happen — but we will do it in one class period!"*

Watch For

Have students read objectives aloud. Pre-teach tectonic plates, compression, and uplift with hand gestures.

Slide 3: Big Question

3 min

Say: *"The fossils on Everest are NOT from a flood. They are from the actual ocean floor. The entire mountain used to be under the sea."*

Watch For

Have students turn and talk: What force could possibly push the ocean floor five miles into the sky?

Slide 4: LEVER Framework

2 min

Say: *"Our model has two things that EARTH controls and three responses that happen over millions of years."*

Watch For

Preview all five components. Emphasize the time scale — these processes are incredibly slow but incredibly powerful.

Slide 5: Activity 1: Components

8 min

Say: *"Can we control how fast tectonic plates move? No — Earth controls that. Can we control how much sediment piles up on the ocean floor? No — nature controls that too."*

Watch For

Guide sorting. Both externals are geological processes humans cannot control. All three internals are results of those processes.

Slide 6: Activity 2: Connections

10 min

Say: *"Notice something special about this model — ALL the relationships are POSITIVE. More plate movement leads to more compression leads to more uplift leads to higher fossils."*

Watch For

Help students draw the chain of positive relationships. Discuss why all positive creates a system where everything amplifies.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch the Fossil Elevation line climb as we run the fast collision scenario. Remember — in real life, this took fifty million years!"*

Watch For

Run scenarios in order. The no-movement scenario shows nothing happening. The fast/thick scenario shows dramatic fossil elevation.

Slide 8: Discoveries

5 min

Say: *"The answer to our mystery: the mountain DID NOT rise up from the land — the OCEAN FLOOR was pushed up to become the mountain!"*

Watch For

Show the geological time sequence: ocean floor with creatures, sediment building, plates colliding, mountain rising with fossils.

Slide 9: STEM Challenge

12 min

Say: *"Build a model that shows a museum visitor exactly how this works — layers of ocean sediment with fossils*

getting pushed up by plate collision."

Watch For

Distribute materials (colored clay/play dough, small shells/beads, cardboard plates). Groups build layered models and simulate collision.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Plate Movement Speed, Sediment Deposit Rate
- **INTERNAL:** Compression Force, Rock Uplift, Fossil Elevation

Relationships

- **Plate Movement Speed to Compression Force:** POSITIVE (+) — Faster plate movement means more energy is transferred when plates collide, creating stronger compression forces. The Indian Plate crashing into Asia at 5 cm/year generates enormous compression that folds and crumples rock layers.
- **Compression Force to Rock Uplift:** POSITIVE (+) — Greater compression force pushes rock layers upward more powerfully. Stronger squeezing means more folding, more crumpling, and faster uplift. This is why rapidly colliding plates create the tallest mountains.
- **Sediment Deposit Rate to Fossil Elevation:** POSITIVE (+) — More sediment accumulation on the ocean floor means thicker layers of fossil-bearing rock. When uplift occurs, thicker layers result in fossils being found at a wider range of elevations. High sedimentation areas produce mountains rich with fossils at many levels.
- **Rock Uplift to Fossil Elevation:** POSITIVE (+) — More uplift pushes the entire rock column higher, carrying embedded fossils to greater elevations. This is the direct mechanism that moves ocean floor fossils from sea level to mountaintop heights.

Key Discoveries

- Tectonic plate collisions create compression forces that squeeze and fold rock, pushing it upward — this is how mountains form
- Ocean sediment layers containing fossils of marine creatures get caught in this uplift and are carried to high elevations over millions of years
- Faster plate movement creates stronger compression, which causes faster rock uplift and higher fossil elevation
- Both plate movement and sediment deposits contribute to fossil elevation — you need thick sediment layers (with fossils in them) AND strong uplift to find ocean fossils on mountaintops

Multiple Choice Post-Assessment Answer Key

1. Scientists find fossils of ocean creatures on top of a mountain that is thousands of feet above sea level. How did those fossils get there?

Correct: C) Tectonic plate collisions pushed ancient ocean floor rock upward over millions of years, carrying the fossils with it

Correct! The rock at the top of mountains like the Himalayas was once ocean floor. Tectonic plates collided and the compression force pushed that ocean floor upward over millions of years, bringing the fossils along.

2. In the model, Plate Movement Speed and Sediment Deposit Rate are external components. What does this mean?

Correct: B) They are the starting conditions that drive the system, and the internal components respond to them

Right! External components are the inputs that drive the system. Plate Movement Speed determines how fast plates collide, and Sediment Deposit Rate determines how thick the fossil layers are. Everything else responds to these.

3. The model shows that Plate Movement Speed has a POSITIVE relationship with Compression Force. What does this mean?

Correct: C) When Plate Movement Speed increases, Compression Force also increases because faster collisions push harder

Exactly! A positive relationship means both go up together. Faster-moving plates crash into each other with more energy, creating stronger compression forces that squeeze and fold rock more powerfully.

4. A student sets Plate Movement Speed to 0% in the model. What happens to all the internal components?

Correct: B) Compression Force drops to zero, Rock Uplift stops, and Fossil Elevation stays where it is

That is right! Without plate movement, there is no collision and no compression force. Without compression, there is no uplift. The fossils stay at whatever elevation they have already reached. Mountains stop growing.

5. The model has two external components that both contribute to where fossils end up. Why do you need BOTH thick sediment layers AND strong plate movement to find many ocean fossils on mountaintops?

Correct: C) Thick sediment layers contain the fossils, and plate movement creates the compression and uplift that pushes those layers to high elevations

Correct! Sediment layers are where the fossils are trapped (more sediment means more fossils). Plate movement creates the compression and uplift that pushes those layers skyward. Without both, you either have no fossils to lift or no force to lift them.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Mountains grow up from the ground like plants"	Mountains do NOT grow from below. They are pushed UP by the horizontal collision of tectonic plates. When two plates crash into each other, the rock has nowhere to go but up. It is like pushing two pieces of paper together on a table — they buckle and rise upward. Mountains are made of rock that was originally at or below the surface and was forced upward by compression.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The ocean used to be much higher and covered the mountains"	The fossils on mountains are NOT from a time when the ocean was higher. The rock itself was UNDER the ocean and was pushed UP by plate movement. The ocean did not drain away — the ocean floor was lifted out of the water. This is proven by the fact that the fossils are embedded deep within the rock layers, not just sitting on the surface.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"This all happened a long time ago and is done"	Mountain building is NOT a thing of the past — it is happening RIGHT NOW. The Himalayas are still growing because the Indian Plate is still pushing into Asia. Mount Everest grows about 4 mm taller every year. Earthquakes in the Himalayan region are caused by the same plate collision that is still building the mountains.